🧮 Age Calculator - Android App Documentation

A modern, clean, and fully functional Android application built with **Kotlin**, **Jetpack Compose (Material 3)**, and **MVVM Architecture**.

Kotlin 2.0+ | Jetpack Compose | Material 3 | MVVM | Min SDK: API 24+ | Java.time Desugaring

🌟 1. Core Features (মূল সুবিধাসমূহ)

- **Exact Age Calculation (নিখুঁত বয়স গণনা):** Calculates age in **Years, Months, and Days** using `java.time.Period` with full leap year accuracy (handling February 29th).
- **Next Birthday Countdown (পরবর্তী জন্মদিনের কাউন্টডাউন):** Displays days remaining until next birthday along with the exact day of the week (e.g., *Friday, Monday*).
- **Birthday Celebration:** Shows an animated `🎂 Happy Birthday! 🎉` banner if today is the user's birthday.
- **Detailed Lifetime Statistics:** Computes Total Days lived, Total Weeks, Total Months, and Total Hours lived.
- **Input Validation:** Prevents picking future dates in the calendar and displays error Snackbars on invalid actions.
- **Modern UI/UX:** Smooth animated card transitions, elevated Material 3 surfaces, and dynamic Light & Dark mode support.

🛠️ 2. Tech Stack & Dependencies (প্রযুক্তি ও লাইব্রেরি)

Technology	Role / Purpose
Kotlin (2.0+)	Official modern Android programming language.
Jetpack Compose (Material 3)	Declarative UI toolkit for building modern, animated, and responsive screens.
MVVM Architecture	Model-View-ViewModel pattern ensuring clean separation of UI and business logic.
StateFlow & Coroutines	Reactive state management for seamless, real-time UI updates.
`java.time` **API**	Modern Java date/time arithmetic (`LocalDate` , `Period` , `ChronoUnit`).
Core Library Desugaring	Enables full `java.time` functionality on devices running Android 7.0+ (API 24+).

📐 3. Architecture & Data Flow (প্রজেক্ট আর্কিটেকচার)

Unidirectional Data Flow (UDF) Architecture:

```
[ AgeCalculatorScreen (UI) ]
        |
        ▼ (User Action: Select Birthdate, Calculate Age, Reset)
[ AgeCalculatorViewModel ] ──► (Validates Date & Calculates Age/Leap-Year/Stats)
        |
        ▼ (Emits new State via StateFlow)
[ UI Re-renders with Smooth Animation ]
```

📁 4. Project Structure & File Guide (ফাইল পরিচিতি)

File Path	Description
`data/AgeCalculatorModel.kt`	Data models: `AgeResult` (calculation results) & `AgeCalculatorUiState` (UI screen state).
`viewmodel/AgeCalculatorViewModel.kt`	Core ViewModel handling business logic, validation, leap year math, and StateFlow.
`ui/components/DatePickerModal.kt`	Material 3 DatePickerDialog modal with disabled future date constraints.
`ui/components/HeroAgeCard.kt`	Primary result card displaying Years, Months, Days badges & Birthday banner.
`ui/components/StatCards.kt`	2×2 Grid displaying Total Days, Weeks, Months, Hours & Next Birthday card.
`ui/AgeCalculatorScreen.kt`	Main Jetpack Compose Scaffold containing the full interactive UI.
`ui/theme/`	Design system tokens: `Color.kt`, `Theme.kt`, and `Type.kt`.
`MainActivity.kt`	Android ComponentActivity entry point with Edge-to-Edge display.

🧮 5. Calculation Algorithms (গাণিতিক ফর্মুলা)

A. Exact Age Calculation (বছর, মাস, দিন)

```
val period = Period.between(birthDate, today)
val years = period.years
val months = period.months
val days = period.days
```

B. Lifetime Unit Conversions (মোট সময়কাল)

```
val totalDays = ChronoUnit.DAYS.between(birthDate, today)
val totalMonths = ChronoUnit.MONTHS.between(birthDate, today)
val totalWeeks = totalDays / 7
val totalHours = totalDays * 24
```

C. Next Birthday Countdown (পরবর্তী জন্মদিন)

```
var nextBdayYear = today.year
var nextBirthdayDate = safeLocalDate(nextBdayYear, birthDate.monthValue, birthDate.dayO

if (nextBirthdayDate.isBefore(today)) {
    nextBdayYear += 1
    nextBirthdayDate = safeLocalDate(nextBdayYear, birthDate.monthValue, birthDate.dayO
}

val daysRemaining = ChronoUnit.DAYS.between(today, nextBirthdayDate)
val dayOfWeek = nextBirthdayDate.dayOfWeek.getDisplayName(TextStyle.FULL, Locale.getDef
```

🚀 6. How to Run in Android Studio

1. Open **Android Studio**.
2. Select **File $\rightarrow$ Open...** and choose `C:\Users\LIMA\Desktop\mubassira`.
3. Wait for **Gradle Sync** to finish.
4. Connect your Android phone (or start an Emulator).
5. Click the green **Run ▶ (Shift + F10)** button to install and run the app.

Generated for Age Calculator Android Project • Built with Kotlin & Jetpack Compose